

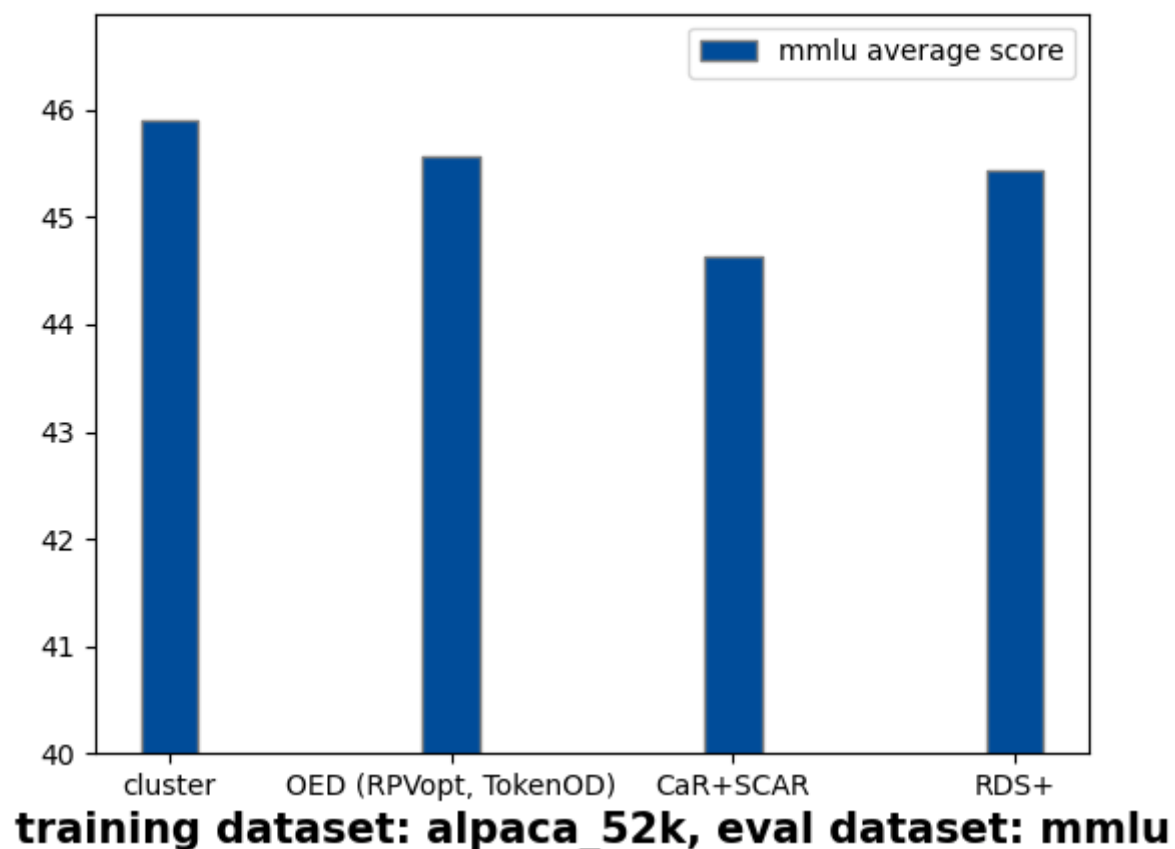
current results

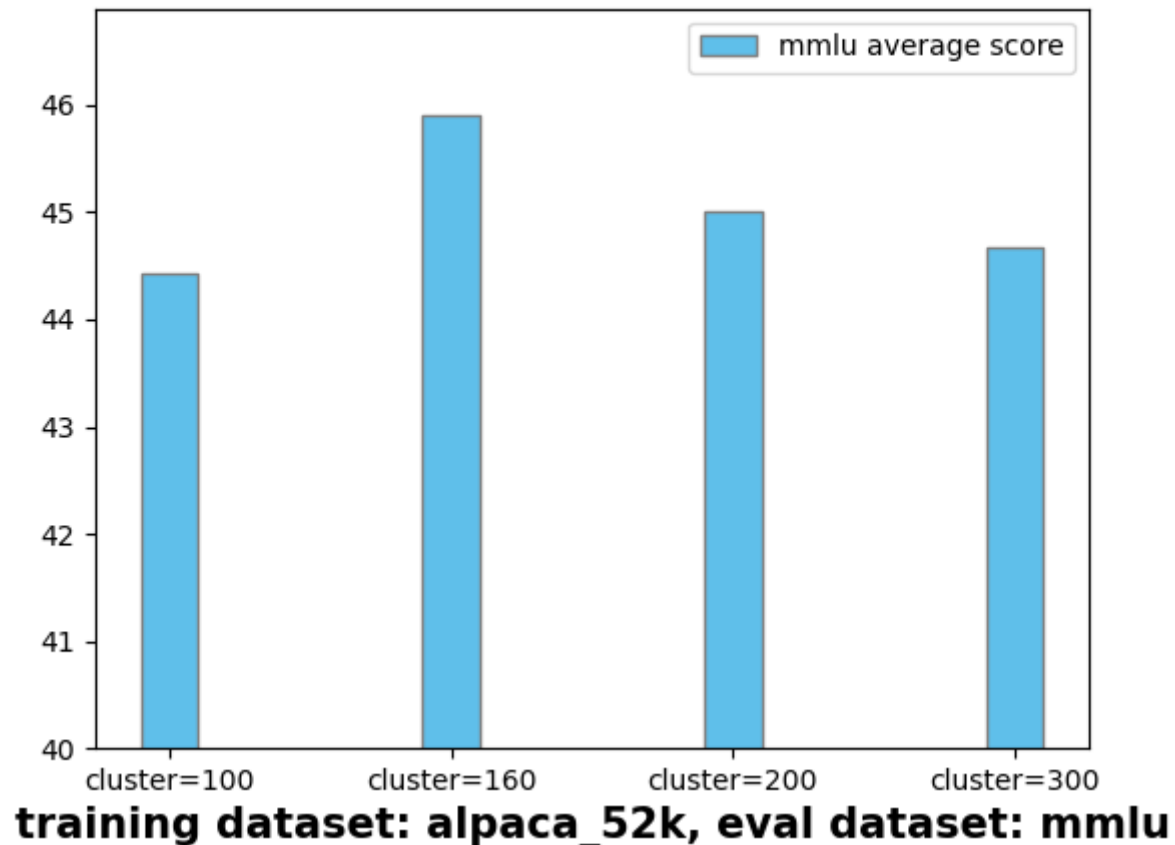
We are currently still in the stage of comparing various methods, and our research focuses on three main categories:

1. Mathematical methods
 - OED + Sketching: TokenOD, RPOpt
 - Clustering: clustering-based sensitivity sampling
2. Score-based (small models): CaR, SCAR
3. Influence-based (high computation cost): RDS+

So far, we have reproduced all methods above. The results are comparable (possibly due to the small model and dataset sizes).

current results:





The mathematical approaches do not require training small models or performing inference on large models needing training, and are computationally more efficient. Therefore, we are trying to preform some naive improvement on math methods like clustering, but we haven't yet decided on a specific direction for deeper investigation.

problems on model-based methods

As for the methods requiring computation on large models, the computational cost is simply too high. Furthermore, these methods have a significant drawback: they require knowledge of both the training dataset and the evaluation dataset simultaneously. They calculate which training data points contribute most to each data point in the evaluation set via model parameters/gradients and select those training points. However, we cannot assume that the distribution of a future evaluation set will be similar to the one used during selection. If we encounter a previously unseen evaluation set with a significantly different distribution (or complex real-world scenarios), the performance might deteriorate significantly.

This issue actually exists for the small model training approach as well. We cannot guarantee that the distribution of the data used during training will be similar to the data used to pre-train the small model. Consequently, the scoring provided by the small model for a training set with a different distribution could become problematic.

problems on clustering-based methods

not stable (sampling)

questions?

1. how can we improve clustering-based method, using different distance metrics or different sampling / clustering methods?
2. sketching method, how can we improve?